Lecture 6

Payment Systems

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Types of Payment Systems

- **Cash**
  - Most common form of payment
  - Instantly convertible into other forms of value
  - No float
- **Checking transfer**
  - Second most common payment form in United States
- **Credit card**
  - Credit card associations
  - Issuing banks
  - Processing centers

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Types of Payment Systems (cont.)

- **Stored value**
  - Funds deposited into account, from which funds are paid out or withdrawn as needed
  - Debit cards, gift certificates
  - Peer-to-peer payment systems
- **Accumulating balance**
  - Accounts that accumulate expenditures and to which consumers make period payments
  - Utility, phone, American Express accounts

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Payment System Stakeholders

- **Consumers**
  - Low-risk, low-cost, refutable, convenience, reliability
- **Merchants**
  - Low-risk, low-cost, irrefutable, secure, reliable
- **Financial intermediaries**
  - Secure, low-risk, maximizing profit
- **Government regulators**
  - Security, trust, protecting participants and enforcing reporting

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E-commerce Payment Systems

- **Credit cards**
  - 42% of online payments in 2013 (United States)
- **Debit cards**
  - 29% online payments in 2013 (United States)
- **Limitations of online credit card payment**
  - Security, merchant risk
  - Cost
  - Social equity:- state of affairs in which all people within a specific society or isolated group have the same status in certain respects, including property rights and equal access to certain social goods and services equal treatment

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How an Online Credit Transaction Works

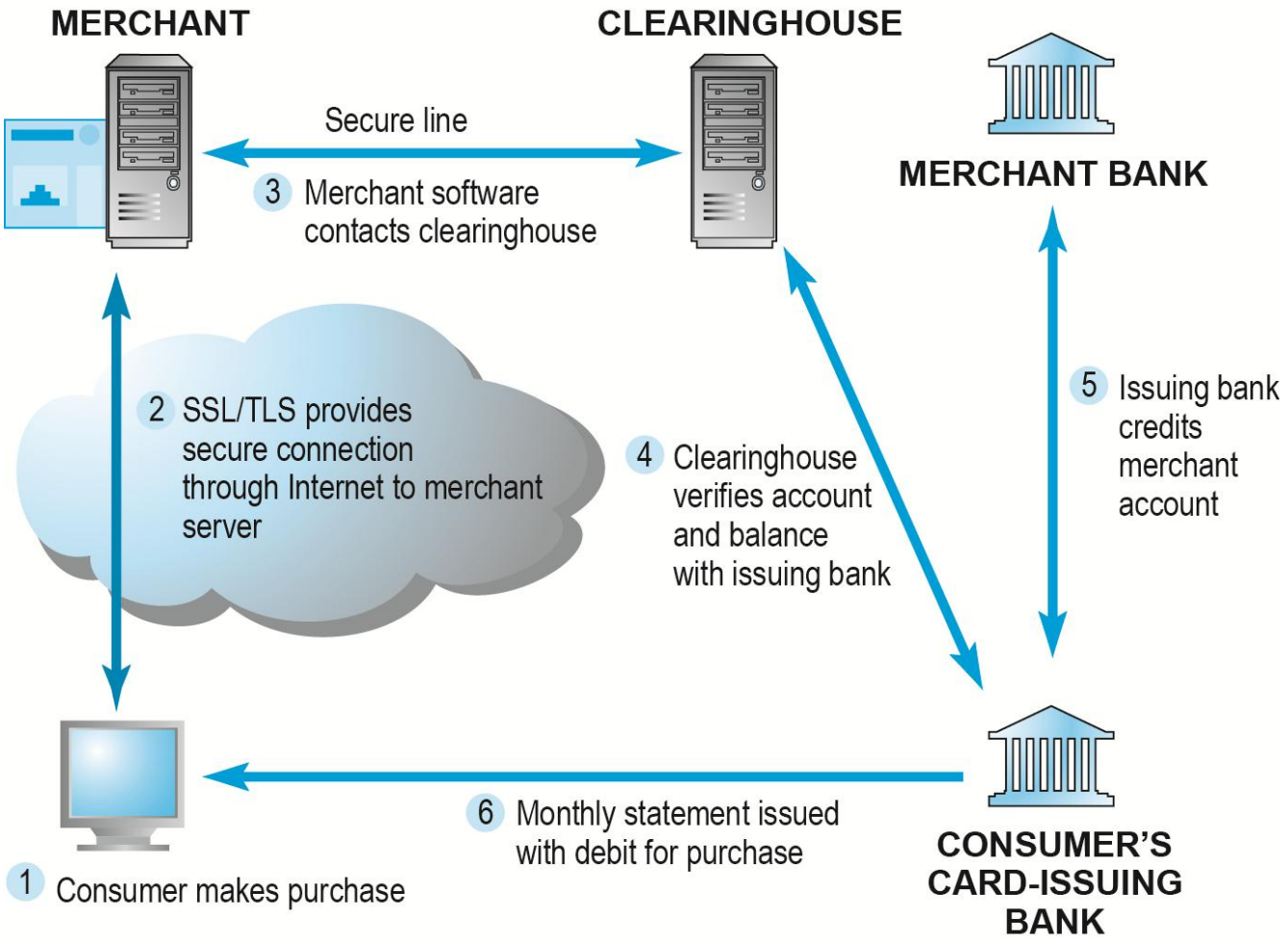


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Alternative Online Payment Systems

- **Online stored value systems:**
  - Based on value stored in a consumer's bank, checking, or credit card account
  - Example: PayPal
- **Other alternatives:**
  - Amazon Payments
  - Google Checkout
  - Bill Me Later
  - WUPay, Dwolla, Stripe

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To make a credit card transaction truly secure and non- refutable, describe sequence of steps that must occur before actual goods, services, or funds flow.

- ***When a consumer wants to buy a certain product, he go to the website and selects the product he wants to buy.***
- ***Once the product is selected, the consumer transaction is moved to the online transaction server where he places an order.***
- ***The informations is exchanged over a secure channel through a private gateway to a processing network.***
- ***The networked banks accept or reject the transaction.***
- ***All this happens in just a matter of seconds.***
- ***eCommerce is a very secure due to the SSL (Secure Socket Layer) technology***
  - *A customer presents his/her credit card information (along with an authenticity signature, or other information such as mother's maiden name) securely to the merchant.*
  - *The merchant validates the customer's identify as the owner of the card account.*
  - *The merchant relays the credit card charge information and signature to its bank or online credit card processing.*
  - *The bank or processing part relays the information to the customer's bank's for authorization approval.*
  - *The Customer's bank returns the credit and the data charge authentication and authorization to the merchant*

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Mobile Payment Systems

- **Use of mobile phones as payment devices established in Europe, Japan, South Korea**
- **Near field communication (NFC)**
  - Short-range (2”) wireless for sharing data between devices
- **Expanding in United States**
  - Google Wallet
    - Mobile app designed to work with NFC chips
  - PayPal
  - Square

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Digital Cash and Virtual Currencies

- **Digital cash**
  - Based on algorithm that generates unique tokens that can be used in "real" world
  - Example: Bitcoin
- **Virtual currencies**
  - Circulate within internal virtual world
  - Example: Linden Dollars in Second Life, Facebook Credits

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Bitcoin

- What are some of the benefits of using a digital currency?
- What are the risks involved to the user?
- What are the political and economic repercussions of a digital currency?
- Have you or anyone you know ever used Bitcoin?

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Electronic Billing Presentment and Payment (EBPP)

- **Online payment systems for monthly bills**
- **50% of all bill payments**
- **Two competing EBPP business models:**
  - Biller-direct (dominant model)
  - Consolidator
- **Both models are supported by EBPP infrastructure providers**

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